Michael Remington Coulter

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PROFESSIONAL EXPERIENCE

Arm | Austin, TX

Senior Software Engineer Apr 2023 - Present

Interconnect Architect | CMN GUI Lead | Tools for internal engineers and external customers.

Recognition: Most Innovative Award, Engineering AI Showcase (Dec 2025), for a project-agnostic multi-agent orchestration and context-gathering system.

- Created all coherent-mesh (CMN) GUI tooling from CMN-800 onward, including Multi-CHI port devices, MCS group configuration, custom routing and mesh deadlock detection interfaces.
- Created the NCI/NoC logical canvas and console for configuration logs and design-rule results; built canvas exports and configuration-report scripts.
- Built a Python NIC-400 XML/CCXML migration engine that reconstructs topology and routes while preserving placement; historical validation recorded 4,130/4,130 valid configurations passing design-rule checks and 5/5 curated invalid configurations rejected.
- Prevented stale or concurrent design-rule responses from updating the wrong view using request IDs and reload checks; added unit and end-to-end regressions.
- Designed a modular AI engineering platform with four project packs, separating shared tooling from product-owned runtimes through manifest-driven configuration.
- Built a Rust/Tauri 2 workbench with React/TypeScript UI, a shared desktop/CLI core, bounded asynchronous Git inspection and verified process controls.
- Built isolated workspaces with dedicated ports, caches and editor state; coordinated concurrent checkout allocation through atomic locks, tokenized leases and ownership-checked recovery.
- Built Jenkins/LSF repair tooling for exact-revision replay and binary patch transfer; required three consecutive clean replays of the latest patch before finalizing flaky-test fixes.
- Built resumable RHEL 8/LSF release automation with archive checks and packaged acceptance, preserving payload hashes/permissions and verifying source/dependency provenance.
- Architected Python Model Context Protocol (MCP) tooling to compile typed design intent into dependency-ordered operations, execute against checkpoints, verify postconditions and support rollback and reproducible Python/YAML exports.
- Built acceptance and bounded agent-evaluation infrastructure for two interconnect construction workflows; recorded results include 5,063 full-suite passes, 92 live checks and exact configuration-hash restoration in a rollback evaluation.

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Arm | Senior Software Engineer | Continued

PROFESSIONAL EXPERIENCE (CONTINUED)

Senior Software Engineer (continued) Apr 2023 - Present

- Introduced Pixi.js to improve NoC canvas performance and created Playwright test infrastructure for newer non-coherent interconnect tooling.
- Implemented grid-aligned physical auto-placement with collision avoidance, grouping and die sizing; tested overlap, containment and adjacency across both renderers.
- Built an Ordered Write Observation wizard for topology, credit/buffer limits and address constraints, with review/apply/restore operations and unit/Playwright tests.
- Delivered guided NoC configuration and logical design comparison; isolated backend sessions to compare topology, routes, attributes and source YAML.

Software Engineer Apr 2020 - Apr 2023

- Implemented Flask services to start and stop Jupyter notebooks and persist files in AWS EFS for Pelion's embedded-ML platform; defined PostgreSQL schemas and built end-to-end tests.
- Built sensor-ingestion and dashboard APIs with InfluxDB, including a fixed-radius neighbor algorithm for workspace proximity detection.
- Designed safety-mechanism RTL for lightweight non-coherent interconnects; used waveform viewers to debug testbenches and RTL/IP during RTL and VIP simulation rotations.

Graduate Software Engineer Jan 2019 - Apr 2020

- Built Verilog FPGA modules and Mbed OS firmware for I2C, GPIO, ADC, DAC and PWM; used waveform viewers to debug testbenches and RTL/IP, and contributed to shield/form-factor PCBs. 2020 Arm Quality Award (team).
- Built Node.js/Express REST APIs, PostgreSQL schemas and React dashboards to monitor occupancy from thermal cameras and PIR sensors.

Earlier Arm internships: IoT Engineering (May-Aug 2018), Systems & Software IP (May-Aug 2017), ATG Verification Enablement (May-Aug 2016), and IT (Oct 2014-Aug 2015).

TECHNICAL SKILLS

Languages: Python, TypeScript/JavaScript, Rust, SQL, Verilog
Applications: Angular, RxJS, Electron, React, Tauri 2, Flask, Node.js/Express, SVG/Pixi
Engineering: MCP, Playwright, Karma, Jenkins, Git, LSF, RHEL 8; interconnects, FPGA, Mbed OS
Cloud / data: Docker, Kubernetes, AWS EKS/EFS, PostgreSQL, InfluxDB, SQLite

EDUCATION

The University of Texas at Austin
Bachelor of Science, Electrical and Computer Engineering | Dec 2018
Primary focus: Computer Architecture and Embedded Systems | Secondary: Integrated Circuits
1st Place (Industry), ECE Capstone Design Award - IoT Wireless Home Security System (2018)